

CYS 339: FRAUD AND DETECTION TECHNOLOGIES

Fraud and Detection Technologies

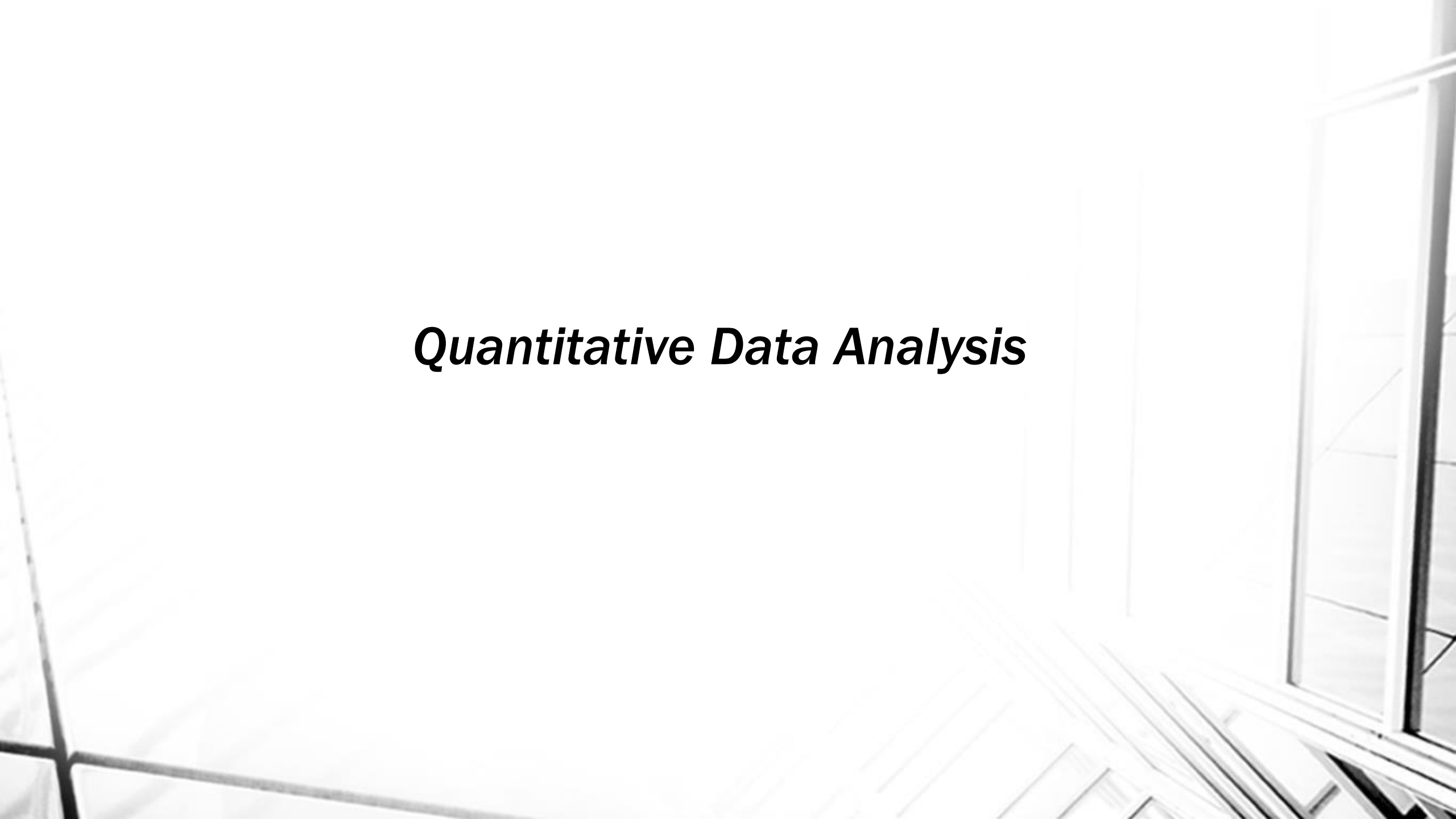
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Session Content

TWO TYPE OF DATA ANALYSIS

1. Quantitative – Sorting, Category, Relationship, Profiling
2. Qualitative – Contextual, Behavioural

Quantitative Data Analysis



Quantitative Data Analysis

- Quantitative data analysis involves numerical data and statistical techniques to detect, analyze, and prevent cyber threats.
- It deals with data which is:

1. Numerical (numbers/numerical values)
2. Measurable (quantified)
3. Comparable (allows for comparisons)

- It is essential for:

1. **Automated threat detection** (e.g., SIEM alerts)
2. **Pattern recognition** (e.g., anomaly detection)
3. **Forensic investigations** (e.g., log analysis)

Types of Quantitative Data Analysis

1. Descriptive Analysis: Summarizes and describes key features of security data

Cybersecurity Applications:

- Calculating average number of daily intrusion attempts
- Reporting monthly malware detection rates
- Summarizing DDoS attack durations

Tools: SIEM dashboards, Excel, Tableau

Example: Our firewall blocks an average of 5,000 malicious packets/hour

2. Diagnostic Analysis: Examines why security events occurred

Cybersecurity Applications:

- Determining root causes of breaches
- Analyzing patterns in successful vs. failed attacks
- Investigating correlations between vulnerability scans and actual exploits

Tools: Splunk, LogRhythm, IBM QRadar

Example: 90% of ransomware incidents originated from phishing emails

Types of Quantitative Data Analysis

3. **Predictive Analysis**: Uses historical data to forecast future threats

Cybersecurity Applications:

- Predicting attack volumes based on historical trends
- Anticipating vulnerability exploitation likelihood
- Forecasting potential DDoS attack times

Tools: Machine learning (Python scikit-learn), Azure Machine Learning

Example: "There's an 85% probability of brute-force attacks on admin accounts this quarter"

Types of Quantitative Data Analysis

4. **Prescriptive Analysis**: Recommends actions based on data insights

Cybersecurity Applications:

- Automated patching prioritization
- Dynamic firewall rule recommendations
- Adaptive authentication measures

Tools: SOAR platforms, Palo Alto Cortex XSOAR

Example: "System recommends immediate isolation of these 3 compromised endpoints

Types of Quantitative Data Analysis

5. Comparative Analysis: Benchmarks security metrics against standards

-Cybersecurity Applications:

- Comparing attack rates across departments
- Measuring security posture against industry benchmarks
- Evaluating effectiveness of new security controls

Tools: SecurityScorecard, CIS Benchmarks

Example: Our mean-time-to-detect (MTTD) is 30% faster than industry average

Types of Quantitative Data Analysis

6. Time Series Analysis: Examines security data points over time

Cybersecurity Applications:

- Detecting seasonal attack patterns
- Identifying slow-and-low attacks
- Monitoring post-incident recovery

Tools: Elasticsearch, Grafana, Pandas

-Example: Noticed 2am-4am is peak time for credential stuffing attacks.

Types of Quantitative Data Analysis

7. Statistical Analysis: Applies mathematical models to security data

Cybersecurity Applications:

- Calculating probabilities of breach scenarios
- Determining confidence intervals for threat detection
- Performing regression analysis on attack vectors

Tools: R, Python (NumPy, SciPy)

Example: There's 99.7% confidence this traffic spike indicates scanning

Types of Quantitative Data Analysis

8. Spatial Analysis: Examines geographic patterns in threats

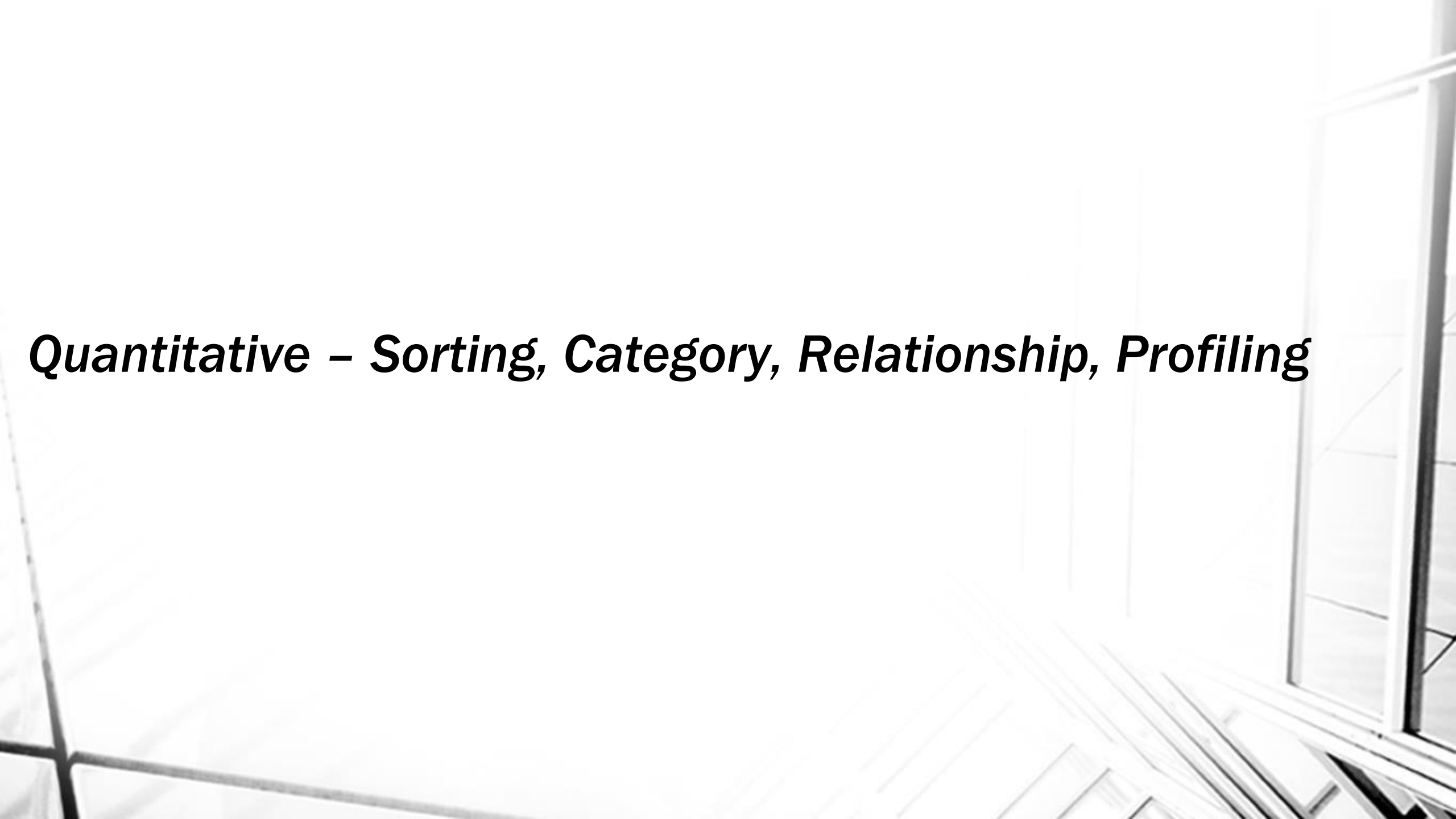
Cybersecurity Applications:

- Mapping attack origins
- Correlating physical locations with breach attempts
- Identifying regional threat clusters

Tools: MaxMind GeoIP, ArcGIS

Example: 65% of our brute-force attacks originate from these 3 countries

Quantitative – Sorting, Category, Relationship, Profiling



Sorting

- Organizing data in a structured order (e.g., chronological, severity-based) to identify trends, outliers, or critical events.
- **Example:** Sorting firewall logs by timestamp to detect brute-force attacks.
  Tool: Splunk (using sort command in SPL) or **ELK Stack** (Kibana's Discover tab).
- **Example:** Prioritizing vulnerabilities by CVSS score.
  Tool: Nessus or **Qualys VMDR** (exports sorted CSV reports).

Sorting: Strengths and weaknesses

Strengths

- **Quick triage** of **high-priority threats**.
- Helps in **regulatory compliance** (e.g., log retention audits).

Limitations

- **False positives** if sorted without context (e.g., benign high-frequency traffic).
- **Static analysis**—does not detect hidden relationships.

Category (Categorization

- Grouping data into predefined classes (e.g., threat types, user roles) for structured analysis.
- **Example:** Classifying malware into ransomware, spyware, or trojans.
  Tool: VirusTotal (automated categorization) or **MITRE ATT&CK** (manual tagging).
- **Example:** Segmenting network traffic (HTTP, DNS, SSH).
  Tool: Wireshark (filter by protocol) or **Zeek** (Bro logs)

Category: Strengths and weaknesses

Strengths

- **Simplifies reporting** (e.g., executive dashboards).
- Enables **automated response** (e.g., blocking all traffic in a malicious category).

Limitations

- **Overgeneralization**—may miss novel attack variants (e.g., zero-day malware).
- **Depends on accurate labels** (misclassification leads to flawed policies)

Relationship (Correlation & Dependency Analysis)

- Identifying statistical or causal links between events (e.g., failed logins + new admin accounts)
- **Example:** Correlating SIEM alerts to detect multi-stage attacks.
  Tool: IBM QRadar (Rule Engine) or **Microsoft Sentinel** (KQL queries).
- **Example:** Linking phishing emails to C2 server IPs.
  Tool: MISP (Threat Intelligence Platform) or **Splunk ES**

Relationship

Strengths

- **Reduces alert fatigue** by grouping related events.
- Uncovers **lateral movement** in APTs (e.g., TTP chaining).

Limitations

- **High computational cost** (large datasets slow down real-time analysis).
- **False correlations** (e.g., coincidental timing ≠ causation)

Profiling (Behavioral Analysis)

- Establishing baselines of normal behavior to detect anomalies (e.g., unusual login times)
- **Example:** Detecting insider threats via user activity logs.
  Tool: Exabeam (UEBA) or **Darktrace** (AI-based anomaly detection).
- **Example:** Network traffic profiling for data exfiltration.
  Tool: Cisco Stealthwatch or **Splunk UBA**

Profiling: Strength and Weaknesses

Strengths

- **Detects zero-day attacks** (behavior-based, not signature-based).
- Adapts to **evolving threats** (machine learning models).

Limitations

- **High false positives** during initial learning phase.
- **Resource-intensive** (requires historical data for training)

Qualitative Data Analysis



Qualitative Data Analysis

- Qualitative analysis focuses on **non-numerical data** (e.g., text, logs, narratives) to understand **meaning, patterns, and motivations** behind cyber threats.
- It prioritizes **depth over breadth**, helping analysts interpret **why** and **how** an attack occurred

Contextual and Behavioral Analysis

Contextual Analysis

- Examining data within its **environmental, situational, or operational context** to derive insights.
- **Contextual analysis** answers ***"Why did this happen?"*** (e.g., phishing lures, social engineering).
- **Example:** Analyzing a phishing email's language (urgency, sender impersonation) to assess legitimacy.

  Tool: Email forensic tools (e.g., MailSniper, PhishTool) parse headers and content.
- **Example:** Investigating a breach by reviewing **SIEM logs + employee interviews** to determine if negligence or malice was involved.

  Tool: Splunk (for log context) + **Case management (e.g., TheHive)**

Contextual Analysis

Strengths

- **Deepens understanding** of attack motives (e.g., hacktivism vs. espionage).
- **Supports legal/incident response** with narrative evidence.

Limitations

- **Time-consuming** (manual review of text/audio/video).
- **Subjective bias** (analyst's interpretation affects outcomes).

Behavioral Analysis

- Studying **human or system behavior patterns** to detect anomalies (e.g., unusual user activity).
- **Behavioral analysis** answers *"How did this happen?"* (e.g., malware actions, user anomalies)
- **Example:** Identifying insider threats by tracking deviations from normal behavior (e.g., sudden access to sensitive files).
  Tool: UEBA platforms (e.g., Exabeam, Darktrace).
- **Example:** Profiling attacker **TTPs (Tactics, Techniques, Procedures)** using **MITRE ATT&CK** framework.
  Tool: Sandboxing (e.g., Cuckoo Sandbox) to observe malware behavior.

Behavioral Analysis

Strengths

- **Detects unknown threats** (no reliance on predefined signatures).
  Adaptive to evolving attacker methods.

Limitations

- **High false positives** (e.g., legitimate but unusual user actions).
  Requires baseline data (initial profiling period needed).

Key Differences: Qualitative vs. Quantitative

Factor	Qualitative	Quantitative
Data Type	Text, logs, narratives	Numbers, metrics, counts
Analysis Focus	"Why" and "How"	"What" and "How much"
Tools	TheHive, MITRE ATT&CK	Splunk, SIEMs, vulnerability scanners
Outcome	Threat intelligence, root causes	Statistics, risk scores

When to Use Qualitative Analysis?

- **Incident Response:** Understanding attacker intent (e.g., ransomware vs. data theft).
- **Threat Hunting:** Profiling adversaries' behavioral patterns.
- **Forensics:** Reconstructing attack timelines from logs + interviews.

Example Workflow:

- **Contextual:** Review phishing email language (urgency, grammar errors).
- **Behavioral:** Analyze attacker's post-breach actions (lateral movement, data accessed)